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From Lab to Field: A Blueprint for a Locally Manufactured, 12-Parameter Soil Sensor to Boost Productivity for Bangladesh's Farmers

This report outlines a comprehensive framework for developing a technically viable, cost-benefit justified, 12-parameter soil testing instrument tailored for use in Bangladesh. The design philosophy prioritizes local manufacturability while strategically incorporating imported components for sophisticated elements. Crucially, the instrument is conceived not as a standalone gadget but as an integrated tool that aligns with the operational realities of Bangladesh's agricultural extension services, aiming to bridge the gap between generalized fertilizer recommendations and precise, site-specific soil management. The analysis synthesizes information on soil degradation challenges, technological feasibility, economic models, and institutional structures within Bangladesh to provide a roadmap for creating a tool that can significantly enhance agricultural productivity, profitability, and sustainability.

Strategic Imperative and Parameter Specification for Bangladeshi Soils

The development of a 12-parameter soil testing instrument in Bangladesh is driven by a profound and multifaceted agricultural crisis rooted in widespread soil degradation and inefficient resource management [24] [29]. The nation’s agricultural foundation is under threat from numerous stressors, including severe land degradation which results in the loss of millions of hectares annually [32], soil erosion in hilly regions, and soil acidification in the north [74]. In the southwest, salinity intrusion has become a major impediment to crop production [37]. These issues are compounded by nutrient imbalances caused by both the over-application of synthetic nitrogen fertilizers, which can deplete other soil nutrients, and the under-application of essential elements like potassium, which limits wheat yields [27] [59]. This complex scenario necessitates a move beyond simplistic, one-

size-fits-all approaches to fertilizer application. The evidence overwhelmingly supports Soil Test Based Fertilizer Application (STFA) or recommendations (STCR) as a superior strategy. Studies have demonstrated that STBA can increase rice grain yield by 20.2% to 32.3% over conventional methods and boost farm profits by up to 30-31% [2] [63]. Furthermore, this precision approach improves nutrient use efficiency, enhances the nutritional quality of crops, and contributes to overall soil health [54] [56] [60]. Recognizing this, government bodies such as the Department of Agricultural Extension (DAE) have been promoting STFAs through official guidelines for years [111144]. However, the impact of these policies is fundamentally limited by the lack of accessible, affordable, and accurate on-site soil testing tools for farmers and extension workers. A 12-parameter instrument is therefore strategically vital; it provides the granular, holistic data required to shift agricultural practices from generalized prescriptions to precise, data-driven decisions tailored to specific soil conditions across Bangladesh's diverse agro-ecological zones [44].

Defining the "12 parameters" requires a hybrid approach that combines directly measurable physical properties with estimable chemical and biological indicators. This strategy balances technological feasibility with scientific relevance, allowing for an integrated device that offers deep diagnostic insight without being prohibitively expensive or complex. The optimal set of parameters should address the core deficiencies identified in Bangladeshi soils and align with national priorities for food security and climate resilience [30] [51]. The following table outlines a proposed set of 12 parameters, categorized by measurement method.

Parameter Category	Specific Parameter	Scientific Rationale & Relevance to Bangladesh
Directly Measurable Physical Properties	pH	Fundamental indicator of nutrient availability; crucial for addressing soil acidification in North Bangladesh [74] [153].
	Electrical Conductivity (EC)	Measures total dissolved salts; essential for monitoring and managing salinity intrusion in Southwest Bangladesh [37].
	Volumetric Water Content (VWC)	Critical for irrigation scheduling and understanding nutrient leaching dynamics in rainfed and irrigated systems [14].
	Temperature	Influences microbial activity and plant growth rates; a basic environmental control for interpreting other measurements [23].
Directly Measurable Chemical Elements	Nitrogen (N), Phosphorus (P), Potassium (K)	The primary macronutrients determining crop yield; direct measurement via electrochemical sensors enables precise fertilizer dosing [46] [69].
	Calcium (Ca), Magnesium (Mg), Sulphur (S)	Secondary nutrients increasingly recognized as limiting factors; sulphur deficiency is a growing concern for crop quality [124][130].
Estimable Chemical/ Biological Indicators	Iron (Fe), Zinc (Zn), Manganese (Mn), Boron (B)	Micronutrient deficiencies hamper crop productivity and contribute to malnutrition [123]; can be estimated via machine learning models [131].
	Soil Organic Matter (SOM) / Carbon (SOC)	Key indicator of long-term soil fertility and carbon sequestration potential; can be correlated with other sensors or estimated via spectroscopy [136][150].
	Redox Potential (Eh)	Measures soil oxidation state; vital for characterizing flooded rice paddy environments and assessing anaerobic conditions [147].

This framework ensures the instrument delivers actionable insights. Direct measurements of pH, EC, moisture, and temperature form the foundational layer, providing immediate context. The inclusion of NPK, Ca, Mg, and S addresses the most pressing yield-limiting nutrient issues. The final four parameters—micronutrients, SOM/SOC, and Eh—are advanced indicators of soil health that are typically only assessable through laboratory analysis. By incorporating estimation algorithms, particularly those leveraging machine learning trained on spectral data (Diffuse Reflectance Spectroscopy - DRS) [150], the instrument can provide a near-lab-level assessment in the field. This approach makes a comprehensive 12-parameter analysis technologically feasible and economically practical, directly empowering farmers and extension agents with the detailed soil intelligence needed to optimize production and steward the land sustainably.

Technical Design and Sensor Integration Strategy

The technical feasibility of the 12-parameter instrument hinges on a modular, open-source hardware architecture combined with robust software for data processing and algorithmic estimation. The core of the device should be a low-cost, high-performance

microcontroller, such as an ESP32, which is prevalent in open-source sensor projects due to its integrated Wi-Fi and Bluetooth capabilities, making it ideal for future IoT connectivity [18] [20]. This central processing unit will manage multiple peripheral sensors, acquire raw data, perform initial calibrations, and run more complex estimation algorithms. The physical design must prioritize durability and ease of use for field agents, featuring a rugged, waterproof casing and a clear, backlit LCD screen for displaying results and guiding users through the testing protocol. Connectivity options like Bluetooth Low Energy (BLE) or LoRaWAN should be integrated to enable wireless data transmission to a companion mobile application or a cloud server, facilitating data aggregation for research and policy purposes [40] [70].

Sensor integration presents the most significant technical challenge, requiring careful selection and rigorous calibration protocols. For the directly measurable parameters, commercially available sensors offer a solid starting point. pH can be measured with a standard glass electrode sensor, while EC is measured with a calomel electrode [151]. Soil moisture is best captured using capacitive sensors like the GS3, which have shown high accuracy when calibrated with second-order polynomial equations, achieving a low Root Mean Square Error (RMSE) of 0.86% in controlled settings [10]. However, it is well-documented that these sensors tend to overestimate moisture content at higher saturation levels [14]. Therefore, the instrument's software must incorporate a dynamic calibration wizard that guides the user to create a unique calibration curve for their specific soil type, a practice validated in multiple studies [13] [20]. Temperature can be measured accurately with inexpensive digital probes [23]. Redox Potential (Eh) is more complex but can be measured with specialized platinum electrodes, providing critical data on the anaerobic status of rice paddies [147].

For the nutrient parameters (N, P, K, Ca, Mg, S), the design must accommodate advanced electrochemical sensors. These sensors can provide rapid, on-site readings but may require importation due to their sophistication [46]. The instrument should feature a modular port system, allowing different sensor cartridges to be swapped depending on the specific analytical needs of the region. This modularity allows for a balance between cost and capability, where a base model might include only NPK, with other modules added as needed. The most innovative aspect of the technical design lies in the estimation of micronutrients (Fe, Zn, Mn, B), organic matter (SOM/SOC), and potentially even residual pesticides or heavy metals [74]. This can be achieved by integrating a small, low-cost Diffuse Reflectance Spectroscopy (DRS) sensor. DRS captures the soil's reflectance spectrum, which contains signatures related to its chemical and mineralogical composition [150]. This spectral data, along with the direct measurements from the physical sensors (pH, EC, moisture, etc.), can serve as input features for a pre-trained Machine

Learning (ML) model. The ML model would then estimate the concentrations of the target parameters. The accuracy of this approach is entirely dependent on the quality of the initial training dataset, which must consist of a wide range of representative Bangladeshi soil samples analyzed simultaneously with the DRS sensor and in a certified laboratory. This fusion of direct sensing and AI-driven estimation represents the most pragmatic path to achieving a true 12-parameter profile in a handheld, field-deployable device.

Economic Viability and Benefit-Cost Justification

A compelling Benefit-Cost Ratio (BCR) is essential for securing investment and ensuring the long-term adoption of the soil testing instrument. The BCR analysis evaluates the present value of all anticipated benefits against the total present value of all associated costs. Given the strong evidence linking soil testing to agricultural gains in Bangladesh, the projected benefits are substantial, likely far outweighing the investment. The World Bank frequently employs NPV and BCR metrics for its projects in the country, underscoring the importance of this financial framework for large-scale initiatives [90].

The cost structure consists of three main components: initial manufacturing and procurement, deployment and training, and ongoing operational and maintenance expenses.

- **Manufacturing Cost:** This is the largest variable. While a baseline price for a multi-parameter sensor exists at approximately BDT 15,199 (~USD 175) [22], a new, integrated 12-parameter device will incur higher costs. A cost-effective strategy involves local manufacturing for non-sophisticated parts (PCBs, enclosures, basic wiring) and importing only high-tech components like the electrochemical sensors and processing units. Leveraging Bangladesh's growing electronics assembly sector can significantly reduce per-unit costs [72] [142].
- **Deployment and Training Costs:** Initial rollout requires funding for the instruments themselves, plus training programs for DAE field agents. This is a standard cost for technology transfer projects and can be integrated into broader agricultural support programs [33] [65].
- **Operational and Maintenance Costs:** These include periodic battery replacement, sensor recalibration kits, and potential repairs. A durable, well-designed device can minimize these recurring expenses.

The benefits are both tangible and intangible, and they accrue to farmers, the environment, and the national economy.

- **Increased Crop Yields and Profitability:** As established, STBA consistently increases yields. A conservative estimate of a 25% yield increase on a typical 2-ton/ha rice farm translates to an additional 500 kg/ha of grain [47] [63]. This directly boosts farm income. One study showed a 31% profit increase over control plots using STB fertilizer application [2]. Another found that treatments using a BAU Soil Testing Kit led to a 61.49% grain yield increase [47]. The combination of higher revenues and optimized inputs leads to improved farmer profitability.
- **Reduced Input Costs:** Precision application enabled by the instrument reduces the wasteful over-application of expensive fertilizers. This can lower fertilizer expenditure by 15-20%, saving hundreds of BDT per hectare annually and mitigating the negative effects of synthetic nitrogen depletion [27].
- **Environmental Benefits:** Reduced fertilizer runoff decreases water pollution and lowers greenhouse gas emissions [27] [146]. Improved soil health enhances carbon sequestration. These environmental gains are aligned with Bangladesh's national climate adaptation goals, which received $3.4 billion in budget allocation for FY 2023/24 [119]. They also contribute to global sustainability targets, such as the UN's SDGs [100].
- **Enhanced Food Security and Nutrition:** Healthier soils produce more nutritious crops, helping to combat malnutrition [54] [56]. This has long-term public health benefits and strengthens national food security [30].

The simplified BCR formula is:

$$\text{BCR} = \frac{\text{Present Value of Total Annual Benefits}}{\text{Initial Investment} + \text{Present Value of Annual Operating Costs}}$$

Given the magnitude of the yield and cost improvements documented in the provided sources, the numerator of this equation is expected to be significantly larger than the denominator. The instrument serves as an investment in human and natural capital, with returns realized through increased agricultural output, enhanced resilience to climate shocks, and a healthier environment. Its alignment with national strategies for climate-smart agriculture further strengthens its case for funding from domestic budgets and international partners like the World Bank, FAO, and UNDP [51] [94] [116].

Alignment with National Agricultural Extension Services

For the soil testing instrument to achieve its intended impact, it must be seamlessly integrated into the workflow of the Department of Agricultural Extension (DAE), the cornerstone of Bangladesh's Agricultural Knowledge and Information System (AKIS) [33]. The AKIS is a pluralistic network, but the public extension service remains pivotal in reaching smallholder farmers [33]. Past reforms, including the New Agricultural Extension Policy (NAEP) of 1996 and the introduction of Common Interest Groups (CIGs), have aimed to make the system more participatory and responsive [33]. However, persistent challenges include poor communication skills among some extension workers and insufficient resources, highlighting the need for effective, user-friendly tools [33]. The proposed instrument must be designed not merely as a data-gathering device but as an advisory aid that empowers DAE agents and enhances their service delivery.

Usability by field agents is paramount. The device must be rugged enough to withstand field conditions, lightweight, and easy to operate. The user interface should feature a simple, multilingual menu system displayed on a clear LCD screen, guiding the agent through the sampling and testing process step-by-step. The entire procedure—from taking a soil sample to receiving a complete 12-parameter report—should take no more than 15-20 minutes. This efficiency is critical for agents who often visit numerous farms in a day. The design must draw lessons from past failures, such as the top-down 'Training and Visit' (T&V) system, which proved ineffective because it did not adequately address farmer needs or empower frontline workers [33]. Instead, the instrument should be positioned as a collaborative tool that facilitates two-way communication and builds the agent's expertise.

Crucially, the instrument must translate raw data into actionable, localized advice compatible with existing national guidelines. The DAE relies on the official Fertilizer Recommendation Guide (FRG) to advise farmers [17]. The instrument's software or its companion mobile app should be able to cross-reference the 12-parameter results with the FRG to generate specific, recommended fertilizer doses for different crops. This transforms the device from a mere measuring tool into an intelligent decision-support system. This functionality directly addresses the knowledge gap among extension workers and ensures that advice is scientifically sound and tailored to the specific soil conditions of a given plot.

Beyond individual farm advice, the instrument offers a powerful opportunity for institutional strengthening and evidence-based policymaking. Equipped with GPS tagging and wireless connectivity, every test result becomes a valuable data point. Aggregated

and anonymized, this data can be transmitted to a central server managed by the DAE or the Bangladesh Agricultural Research Council (BARC) [144]. This creates a real-time, high-resolution map of soil health across the country, allowing officials to identify emerging problems like acidification or salinization hotspots, track the effectiveness of interventions, and update national fertilizer recommendations accordingly [44] [75]. This capacity for continuous monitoring and data-driven adaptation was a key weakness in previous extension efforts [33]. By turning every test into a data contribution, the instrument helps the DAE evolve from a service provider into a proactive knowledge hub, ultimately improving the quality and relevance of agricultural support nationwide.

Local Manufacturability and Supply Chain Strategy

A cornerstone of the project's sustainability and affordability is a balanced supply chain strategy that maximizes local manufacturing while strategically sourcing sophisticated components internationally. This approach leverages Bangladesh's growing industrial capabilities, supports local economic development, and mitigates risks associated with import dependency and currency fluctuations. The success of this strategy depends on a clear understanding of the country's manufacturing landscape and the specific skill sets required for assembling a high-tech agricultural device.

Bangladesh has made significant strides in its electronics manufacturing sector, transitioning from garments to volume electronics assembly and developing smallholder farming technologies [53] [72]. This provides a strong foundation for producing many of the instrument's non-specialized components. The plan should explicitly identify Bangladeshi suppliers for printed circuit boards (PCBs), injection-molded plastic casings, batteries, wiring harnesses, and basic electronic components like resistors and capacitors. Establishing partnerships with local manufacturers not only reduces costs by avoiding import duties and lengthy logistics chains [108] but also creates skilled jobs and fosters a domestic ecosystem for agricultural technology. The country's Technical and Vocational Education and Training (TVET) sector, though facing challenges, is a critical source of talent [9] [99]. Collaborating with institutions like the Bangladesh University of Engineering and Technology (BUET) could help develop specialized curricula and ensure a pipeline of technicians capable of handling PCB design, component sourcing, and quality assurance [65] [142].

However, certain components demand a level of precision and miniaturization currently beyond the reach of local production. These "sophisticated elements" must be sourced internationally to ensure the instrument meets its performance specifications. This list includes:

- **High-Precision Electrochemical Sensors:** The sensors for N, P, K, and potentially other nutrients require specialized membranes and electrodes that are best produced by established sensor manufacturers globally.
- **Precision Measurement Probes:** High-quality glass pH electrodes and calomel EC probes are critical for accuracy and are largely imported [151].
- **Microcontroller and Processing Unit:** While microcontrollers are widely available, the specific high-performance chip chosen for the central processing unit might be imported.
- **Advanced Components:** Any integrated Diffuse Reflectance Spectroscopy (DRS) module or specialized redox (Eh) sensor would fall into this category.

The supply chain must be resilient and transparent. A dual-sourcing strategy for critical imported components can mitigate geopolitical or logistical disruptions. Furthermore, maintaining a stock of spare parts and calibration solutions within Bangladesh is essential for after-sales service and maintenance, reducing downtime for end-users. The involvement of private companies, alongside public agencies like the DAE, is crucial for navigating this complex supply chain, as the private sector often has more experience in global procurement and logistics [33]. Ultimately, the goal is to create a virtuous cycle: initial reliance on imports for key components funds the development of local manufacturing capacity, which in turn lowers the overall cost of the instrument and strengthens the national agricultural technology sector.

Implementation Roadmap and Strategic Recommendations

The successful development and deployment of a 12-parameter soil testing instrument for Bangladesh requires a phased, collaborative, and evidence-based implementation roadmap. Moving from concept to a widely adopted tool demands careful planning that integrates technical development, economic analysis, institutional engagement, and market readiness. The following strategic recommendations outline a logical sequence of actions to guide the project toward its objectives.

First, establish a multidisciplinary project consortium. This team must bring together experts from various fields: soil scientists to define parameter relevance and ground-truth data; sensor engineers and data scientists to design the hardware and develop the estimation algorithms; agricultural economists to refine the BCR model; and representatives from the Department of Agricultural Extension (DAE) and the Bangladesh Agricultural Research Council (BARC) to ensure alignment with national priorities and extension workflows [111][144].

Second, conduct a comprehensive pilot study to build a foundational database. Before any device can be built, a robust dataset correlating lab-measured soil properties with sensor outputs is required. This involves collecting a statistically significant number of soil samples from across Bangladesh's diverse agro-ecological zones [44]. Each sample must be analyzed in a certified laboratory for all 12 target parameters. Simultaneously, the same samples should be tested with prototype sensors (e.g., capacitance probes, DRS sensors) to establish the calibration curves and train the machine learning models. This ground-truthing phase is the most critical and time-consuming step, but its quality will determine the ultimate accuracy and reliability of the final product.

Third, develop a Minimum Viable Product (MVP). Rather than attempting to build the full 12-parameter version immediately, start with an MVP that incorporates the most critical and easily measurable parameters: pH, EC, Temperature, Moisture, and NPK. This allows the team to validate the core hardware-software integration, test the user interface with a small group of DAE agents, and demonstrate the value proposition in a real-world setting. Feedback from the MVP phase will be invaluable for refining the design before committing to mass production.

Fourth, engage the DAE early and continuously. The DAE is not just a potential end-user but a key partner in the entire lifecycle of the project. Involve them from the outset in defining the instrument's features, particularly its advisory software. Co-design the workflow so that the device integrates seamlessly into their existing systems and reporting requirements. This partnership will be instrumental in overcoming adoption barriers and ensuring the tool becomes a standard part of their toolkit. Addressing known weaknesses in extension services, such as communication gaps, should be a primary design consideration [33].

Finally, pursue strategic funding and scale-up. The project's proposal should be framed around its high BCR, its alignment with national climate-smart agriculture goals [51], and its potential to create local jobs and strengthen the domestic electronics industry. Target funding from international organizations with a strong presence in Bangladesh, such as the World Bank, the Food and Agriculture Organization (FAO), and the United Nations

Development Programme (UNDP), which support initiatives in climate change adaptation, sustainable agriculture, and private sector development [1] [94] [116]. A phased rollout, starting with a targeted deployment in specific districts, will allow for monitoring, evaluation, and necessary adjustments before a nationwide launch. By following this roadmap, the project can systematically navigate the path from a promising concept to a transformative tool for Bangladeshi agriculture.

Reference

1. [PDF] World Bank Document https://documents1.worldbank.org/curated/en/099040523193537513/pdf/P17637402389fa0e0b1dd0012aab847e8f.pdf
2. (PDF) Impacts of soil nutrient management practices on soil fertility ... https://www.researchgate.net/publication/350120277_Impacts_of_soil_nutrient_management_practices_on_soil_fertility_nutrient_uptake_rice_Oryza_sativa_L_productivity_and_profitability
3. Evidence from fertilizer quantity recommendations in Bangladesh https://www.sciencedirect.com/science/article/abs/pii/S0304387823001517
4. Soil Test Based Fertilizer Application Improves Productivity ... - MDPI https://www.mdpi.com/2073-4395/11/9/1756
5. Evaluation of different fertilizer management guidelines for boro rice ... https://www.academia.edu/106240749/Evaluation_of_different_fertilizer_management_guidelines_for_boro_rice_cultivation_in_south_central_coastal_region_of_Bangladesh
6. [PDF] Bangladesh - FAO Knowledge Repository https://openknowledge.fao.org/server/api/core/bitstreams/fc5d537e-a251-4940-8074-016efb453433/content
7. [PDF] Comparative performance of different fertilizer Recommendation ... https://sdiopr.s3.ap-south-1.amazonaws.com/2024/Apr/17-Apr-24/2024_AJRCS_113880/Rev_AJRCS_113880_Nur_A.pdf
8. 49310-002: Yunnan Lincang Border Economic Cooperation Zone ... https://www.adb.org/projects/49310-002/main
9. [PDF] Availability of Data related to Technical and Vocational Education ... https://www.ilo.org/sites/default/files/wcmsp5/groups/public/@asia/@ro-bangkok/@ilo-dhaka/documents/publication/wcms_106496.pdf

10. Calibration and Assessment of Capacitance-Based Soil Moisture ... https://www.researchgate.net/publication/339414132_Calibration_and_Assessment_of_Capacitance-Based_Soil_Moisture_Sensors
11. Review of Novel and Emerging Proximal Soil Moisture Sensors for ... https://pmc.ncbi.nlm.nih.gov/articles/PMC7730258/
12. (PDF) Performance assessment of soil moisture sensors under ... https://www.researchgate.net/publication/348153777_Performance_assessment_of_soil_moisture_sensors_under_controlled_conditions_in_laboratory_setting_and_recommendations_for_field_deployment
13. (PDF) Calibration of a soil moisture sensor with disturbed and ... https://www.researchgate.net/publication/328057292_Calibration_of_a_soil_moisture_sensor_with_disturbed_and_undisturbed_soil_samples_from_Bahia
14. FA3003-1) in automated drip irrigation for upland, rainfed farms https://www.researchgate.net/publication/398581162_Quantifying_calibration_strategies_for_capacitive_soil-moisture_sensors_V12_SKU_FA3003-1_in_automated_drip_irrigation_for_upland_rainfed_farms
15. (PDF) Farm Machinery Testing and Standardization in Bangladesh https://www.researchgate.net/publication/391670235_Farm_Machinery_Testing_and_Standardization_in_Bangladesh_A_Policy_Review
16. [PDF] Bangladesh national Conservation strategy 2021-2036 https://faolex.fao.org/docs/pdf/BGD219210.pdf
17. (PDF) Prospect of Developing Soil Health Index in Bangladesh https://www.academia.edu/107540244/Prospect_of_Developing_Soil_Health_Index_in_Bangladesh
18. Development and Validation of a Low-Cost and Open-Source Soil ... https://www.mdpi.com/1424-8220/25/22/6907
19. (PDF) picoSMMS: Development and Validation of a Low-Cost and ... https://www.researchgate.net/publication/397603574_picoSMMS_Development_and_Validation_of_a_Low-Cost_and_Open-Source_Soil_Moisture_Monitoring_Station
20. Designing and field calibration of low-cost microcontroller-based soil ... https://pmc.ncbi.nlm.nih.gov/articles/PMC12528689/
21. Development of low-cost handheld soil moisture sensor for farmers ... https://www.frontiersin.org/journals/environmental-science/articles/10.3389/fenvs.2025.1590662/full

22. Soil Sensor RS485 5 Pin PH NPK Storage Temperature And ... https://store.roboticsbd.com/soil-waterph/2300-soil-sensor-rs485-5-pin-ph-npk-temperature-and-humidity-ec-sensor-conductivity-tester-multi-parameter-robotics-bangladesh.html?srsltid=AfmBOoqm9vjMVz6piGWD2P5ol0EDLmtd114_kw1CZCRQWZtdcUmXtOe7
23. 2026 Soil Sensors - Atech Sensor Co., Ltd - Page 1 https://atech-sensor.en.made-in-china.com/product-group/nqSTrfeCYLVP/Soil-Sensors-catalog-1.html
24. [PDF] Status of the - World's Soil Resources - FAO.org https://www.fao.org/fileadmin/user_upload/newsroom/docs/FAO-world-soils-report-SUMMARY.pdf
25. Review of low-cost, off-grid, biodegradable in situ autonomous soil ... https://www.sciencedirect.com/science/article/pii/S016816992400680X
26. [PDF] 2021. The impact of disasters and crises on agriculture and food ... https://openknowledge.fao.org/server/api/core/bitstreams/30c0d98d-1c21-48ef-b5d9-8d988e6fa6f2/content
27. [PDF] Agricultural Pollution - Main References - World Bank Document https://openknowledge.worldbank.org/bitstreams/4cbfb308-f821-5977-a314-5c0ec46403db/download
28. [PDF] Status of the World's Soil Resources - FAO Knowledge Repository https://openknowledge.fao.org/server/api/core/bitstreams/6ec24d75-19bd-4f1f-b1c5-5becf50d0871/content
29. Soil Degradation, Land Scarcity and Food Security - MDPI https://www.mdpi.com/2071-1050/8/3/281
30. [PDF] Strengthening Strategic Grain Reserves to Enhance Food Security https://documents1.worldbank.org/curated/en/099042625211562573/pdf/P504545-488431b2-0565-40f9-852c-e8db32d22559.pdf
31. [PDF] A/74/238 - General Assembly - the United Nations https://docs.un.org/en/a/74/238?direct=true
32. Rethinking Global Soil Degradation: Drivers, Impacts, and Solutions https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025RG000883
33. agricultural extension approach for bangladesh based on field ... https://www.academia.edu/27824474/AGRICULTURAL_EXTENSION_APPROACH_FOR_BANGLADESH_BASED_ON_FIELD_EXPERIENCES
34. Digital Agriculture Innovations in Bangladesh: A Situational Analysis ... https://onlinelibrary.wiley.com/doi/abs/10.1002/tie.22421
35. (PDF) Understanding the determinants and impact of the adoption of ... https://www.researchgate.net/publication/365822575_Understanding_the_determinants_and_impact_of_the_adoption_of_technologies_for_sustainable_farming_systems_in_water-scarce_areas_of_Bangladesh

36. Rice bran and rice bran oil production perspective in Bangladesh: a ... https://scijournals.onlinelibrary.wiley.com/doi/10.1002/jsfa.14104?af=R
37. Impact of salinity intrusion on agriculture of Southwest Bangladesh ... https://www.researchgate.net/publication/341432509_Impact_of_salinity_intrusion_on_agriculture_of_Southwest_Bangladesh-A_review
38. (PDF) IOT-Based Smart Agriculture in Bangladesh: An Overview https://www.researchgate.net/publication/378968426_IOT-Based_Smart_Agriculture_in_Bangladesh_An_Overview
39. Applications of internet of things (IoT) and sensors technology to ... https://www.sciencedirect.com/science/article/pii/S2090447923003982
40. Innovations in Sensor-Based Systems and Sustainable Energy ... https://www.mdpi.com/2673-8392/5/2/67
41. Integration of IoT, Machine Learning, and Sensors for Intelligent ... https://onlinelibrary.wiley.com/doi/10.1155/jcnc/6611890
42. Use of artificial intelligence for precision agriculture in Bangladesh https://www.researchgate.net/publication/361092694_Use_of_artificial_intelligence_for_precision_agriculture_in_Bangladesh
43. Trends of Soil and Solution Nutrient Sensing for Open Field ... - MDPI https://www.mdpi.com/1424-8220/25/2/453
44. Spatial Variability and Geostatistical Analysis of Soil Properties in ... https://onlinelibrary.wiley.com/doi/10.1155/2021/6639180
45. Sustainability-driven fertilizer recommender system for coffee crops ... https://www.frontiersin.org/journals/sustainable-food-systems/articles/10.3389/fsufs.2024.1445795/full
46. Development of electrochemical sensors for quick detection of ... https://pubs.rsc.org/en/content/articlehtml/2024/ra/d4ra00034j
47. Comparative Performance of Different Fertilizer Recommendation ... https://www.researchgate.net/publication/379829284_Comparative_Performance_of_Different_Fertilizer_Recommendation_Methods_on_Growth_and_Yield_of_Rice_Oryza_sativa_in_Old_Brahmaputra_Floodplain_Soils
48. [PDF] MS Islam - Soil Science Society of Bangladesh https://sssb1958.com/wp-content/uploads/2025/09/1.-MS-Islam.pdf
49. Agronomic, economic, and environmental performance of nitrogen ... https://www.sciencedirect.com/science/article/pii/S0378429018312760

50. [PDF] Evaluation of FAO's Contribution to Bangladesh https://openknowledge.fao.org/bitstreams/f1a10629-7275-4d8e-929a-8bf6a2e848c2/download
51. [PDF] bangladesh climate-smart agriculture investment plan https://documents1.worldbank.org/curated/en/936881574884000754/pdf/Bangladesh-Climate-Smart-Investment-Plan-Investment-opportunities-in-the-agriculture-sector-s-transition-to-a-climate-resilient-growth-path.pdf
52. https://www.fao.org/family-farming/xls/ru/?res_typ... https://www.fao.org/family-farming/xls/ru/?res_type=53206&country_iso3=BGD
53. [PDF] Promoting Agri-Food Sector Transformation in Bangladesh https://documents1.worldbank.org/curated/en/475301590995321523/pdf/Bangladesh-Promoting-Agri-Food-Sector-Transformation-in-Bangladesh-Policy-and-Investment-Priorities.pdf
54. [PDF] Technical guidelines on - soilsfor nutrition https://openknowledge.fao.org/server/api/core/bitstreams/daf25621-7a13-461e-9fe5-bdaf6b04ac6a/content
55. [PDF] Climate-Smart Agriculture Sourcebook https://www.fao.org/3/i3325e/i3325e.pdf
56. [PDF] Collecting the Dirt on Soils - World Bank Open Knowledge Repository https://openknowledge.worldbank.org/server/api/core/bitstreams/3c33f163-a5c2-53e7-81a0-a1862d235824/content
57. [PDF] Climate Change Mitigation Finance for Smallholder Agriculture https://www.fao.org/4/i2485e/i2485e00.pdf
58. [PDF] PROCUREMENT PLAN (Textual Part) - World Bank Document https://documents1.worldbank.org/curated/en/099031325022042673/pdf/P161534-1f8dcf1a-192c-4575-8f78-b95d6b3515ad.pdf
59. Nutrient management for increased productivity of rice–wheat ... https://www.sciencedirect.com/science/article/abs/pii/S0378429005002029
60. (PDF) Soil Test Based Fertilizer Application Improves Productivity ... https://www.researchgate.net/publication/354248080_Soil_Test_Based_Fertilizer_Application_Improves_Productivity_Profitability_and_Nutrient_Use_Efficiency_of_Rice_Oryza_sativa_L_under_Direct_Seeded_Condition
61. Response of grain yield and soil health to the individual application ... https://link.springer.com/article/10.1007/s44279-025-00201-y
62. Effect of Missing Nutrient Elements on Grain Yield of Wet Season ... https://www.scirp.org/journal/paperinformation?paperid=92079
63. (PDF) Soil Test Based Fertilizer Application Improves Productivity ... https://www.academia.edu/58788594/

Soil_Test_Based_Fertilizer_Application_Improves_Productivity_Profitability_and_Nutrient_Use_Efficiency_of_Rice_Oryza_sativa_L_under_Direct_Seeded_Condition

64. Modification of Nutrient Requirements for a Four Crop-Based ... - MDPI https://www.mdpi.com/2071-1050/14/12/7194
65. [PDF] supply assessment in the Chattogram–Cox's Bazar region ... https://www.ilo.org/sites/default/files/2024-04/Bangladesh-Skills-Assessment-Chattogram-CoxsBazar-Narrative-April-2024-en.pdf
66. [PDF] IDA Sustainable Development Bonds | FY24 Impact Report https://thedocs.worldbank.org/en/doc/c1777be525a71b9d21c9cdd46471b81d-0340022025/original/IDA-Impact-Report-Project-Table.pdf
67. Identification and analysis of key factors limiting the performance of ... https://www.sciencedirect.com/science/article/pii/S0168169925005861
68. Soil Nutrient Monitoring Technologies for Sustainable Agriculture https://www.mdpi.com/2071-1050/17/18/8477
69. (PDF) Conventional to Modern Methods of Soil NPK Sensing https://www.researchgate.net/publication/376020116_Conventional_to_Modern_Methods_of_Soil_NPK_Sensing_A_Review
70. Internet of Things and Wireless Sensor Networks for Smart ... https://ieeexplore.ieee.org/iel7/6287639/10005208/10371307.pdf
71. Sustainable aquaculture: An Iot-integrated system for real-time water ... https://www.sciencedirect.com/science/article/pii/S0144860925001098
72. That 99% figure is not a dream anymore - LinkedIn https://www.linkedin.com/pulse/99-figure-dream-anymore-rizwanul-islam-uzhyc
73. Open-Source Photosynthetically Active Radiation Sensor for ... - MDPI https://www.mdpi.com/2079-9292/14/11/2225
74. [PDF] National Food and Nutrition Security Policy https://faolex.fao.org/docs/pdf/BGD214404.pdf
75. Assessment of the soil fertility index (SFI) for sustainable nutrient ... https://link.springer.com/article/10.1007/s44378-025-00130-8
76. [PDF] Good agricultural practices (GAP) - FAO Knowledge Repository https://openknowledge.fao.org/server/api/core/bitstreams/7dcb36a8-b206-41bc-948f-b1edff4c1d06/content
77. Regenerating rural soil and ecosystems: A 15-year systematic ... https://www.sciencedirect.com/science/article/pii/S0048969725015669
78. [PDF] Baseline study of biosaline agriculture and roadmap to concerted ... https://openknowledge.fao.org/server/api/core/bitstreams/90dbf25e-73cd-4280-b01f-0154e6969416/content

79. [PDF] pest management plan (pmp) - World Bank Document https://documents1.worldbank.org/curated/en/099135111242211399/pdf/P1714620b2fe080a80a40f0633aeb1e1730.pdf
80. How productive are rice farmers in Sri Lanka? The impact of ... - PMC https://pmc.ncbi.nlm.nih.gov/articles/PMC8255234/
81. Regenerative Agriculture—A Literature Review on the Practices and ... https://www.mdpi.com/2071-1050/15/3/2338
82. [PDF] World Without End: Economics, Environment, and Sustainable ... https://documents1.worldbank.org/curated/en/716131468764410726/pdf/multi0page.pdf
83. [PDF] State of knowledge of soil biodiversity https://www.cbd.int/doc/c/fdb6/6467/1fed250d39261056b911d813/sbstta-24-inf-08-en.pdf
84. Using Field-Based Monitoring to Enhance the Performance ... - MDPI https://doi.org/10.3390/w12123453
85. Indoor air quality monitoring and source apportionment using low ... https://iopscience.iop.org/article/10.1088/2515-7620/ad1cad
86. Accurate Water Level Monitoring in AWD Rice Cultivation Using ... https://arxiv.org/html/2412.08477v1
87. [PDF] Smart Crop Cultivation System Using Automated Agriculture ... https://pdfs.semanticscholar.org/3629/fd3712b6b3dccdcd2ffbdf88c54e93d3a821.pdf
88. Comprehensive analysis of methods for estimating actual paddy ... https://www.frontiersin.org/journals/water/articles/10.3389/frwa.2025.1553732/full
89. English Text (337.06 KB) - Open Knowledge Repository - World Bank https://openknowledge.worldbank.org/bitstreams/7ad22a18-375c-4abe-80d4-eba773bb2e77/download
90. English Text (304.01 KB) - Open Knowledge Repository https://openknowledge.worldbank.org/bitstreams/70c4f868-3bc1-5522-848d-5d645aaba235/download
91. [PDF] RESTORING STABILITY, REBUILDING TRUST https://documents1.worldbank.org/curated/en/099120725140521030/pdf/P501730-eb5acf1a-8ebe-45d8-847c-7c2bd17a2e52.pdf
92. [PDF] bangladesh's third nationally determined contribution (ndc 3.0) | unfccc https://unfccc.int/sites/default/files/2025-09/Bangladesh%20Third%20Nationally%20Determined%20Contribution%20%28NDC%203.0%29.pdf
93. [PDF] Repurposing Agricultural Support: 2025 Policy Compendium https://thedocs.worldbank.org/en/doc/7cc1a0b005a879c4b2c3efd5f72da444-0320012025/original/2025-compendium.pdf

94. [PDF] Bangladesh: agriculture sector - UNDP Climate Promise https://climatepromise.undp.org/sites/default/files/research_report_document/undp-iff-bangladesh-assessment-agriculture-en.pdf
95. [PDF] Evaluation of the FAO Flexible Voluntary Contribution Mechanism https://openknowledge.fao.org/server/api/core/bitstreams/2b4ef5d7-e760-42de-a3a8-3124bcf17f00/content
96. (PDF) Soil-Test-Based Optimum Fertilizer Doses for Attaining Yield ... https://www.researchgate.net/publication/281856558_Soil-Test-Based_Optimum_Fertilizer_Doses_for_Attaining_Yield_Targets_of_Rice_under_Midland_Alfisols_of_Eastern_India
97. Wheat and maize-derived water-washed and unwashed biochar ... https://www.sciencedirect.com/science/article/abs/pii/S0301479721013128
98. [PDF] J Bangladesh Agril Univ 19(2): 168–177, 2021 - CABI Digital Library https://www.cabidigitallibrary.org/doi/pdf/10.5555/20220203716
99. (PDF) Exploring Historical Background of TVET Sector in Bangladesh https://www.researchgate.net/publication/380576601_Exploring_Historical_Background_of_TVET_Sector_in_Bangladesh
100. the 2030 Agenda for Sustainable Development - IDEAS/RePEc https://ideas.repec.org/r/ess/wpaper/id7559.html
101. Terrorism and Transatlantic Relations Threats and Challenges https://link.springer.com/content/pdf/10.1007/978-3-030-83347-3.pdf
102. Contemporary Art Biennials - Our Hegemonic Machines in Times of ... https://www.academia.edu/43721618/Contemporary_Art_Biennials_Our_Hegemonic_Machines_in_Times_of_Emergency
103. [PDF] FAO FERTILIZER AND PLANT NUTRITION BULLETIN https://www.fao.org/3/a0443e/a0443e.pdf
104. 122934-WP-P153343-PUBLIC-Vietnam-crops-ENG.txt https://documents1.worldbank.org/curated/en/988621516787454307/txt/122934-WP-P153343-PUBLIC-Vietnam-crops-ENG.txt
105. Degraded land: Assessment, impact, restoration, and challenges https://www.sciencedirect.com/science/article/pii/S0301479725044470?dgcid=rss_sd_all
106. [PDF] IAEA TECDOC SERIES https://www-pub.iaea.org/MTCD/Publications/PDF/TE_1784_web.pdf
107. [PDF] The-causes-nature-and-rate-of-soil-degradation-in-the-northernmost ... https://documents1.worldbank.org/curated/en/959831493305706913/pdf/The-causes-nature-and-rate-of-soil-degradation-in-the-northernmost-states-of-Nigeria-and-an-assessment-of-the-role-of-fertilizer-in-counteracting-the-processes-of-degradation.pdf

108. [PDF] TRADE POLICY FRAMEWORK UPDATE https://faolex.fao.org/docs/pdf/van214216.pdf
109. [PDF] 46625 - World Bank Documents & Reports https://documents1.worldbank.org/curated/en/994361468159335243/pdf/466250NWP0Box31inal0June01302008111.pdf
110. multi-page.txt - World Bank Documents & Reports https://documents1.worldbank.org/curated/en/184521468743408092/txt/multi-page.txt
111. Agrometeorological Information for Climate Resilient Agriculture in ... https://wmo.int/media/magazine-article/agrometeorological-information-climate-resilient-agriculture-bangladesh
112. [PDF] ANNEX 1 Using the Programme Results Framework from the Project ... https://mptf.undp.org/sites/default/files/documents/2025-05/all_annexes.pdf
113. [PDF] bangladesh - Green Policy Platform https://info.undp.org/docs/pdc/Documents/BGD/SDGs-Bangladesh_VNR_2020.pdf
114. [PDF] Financing the UN Development System September 2025 https://mptf.undp.org/sites/default/files/documents/2025-09/_financing_the_unds_report_2025_0.pdf
115. [PDF] One Future - UNDP https://www.undp.org/sites/g/files/zskgke326/files/2025-11/cop_publication_0.pdf
116. [PDF] FINAL EVALUATION REPORT https://erc.undp.org/evaluation/documents/download/24828
117. [PDF] Umbrella Programme to Support Develop https://www.undp.org/sites/g/files/zskgke326/files/2024-10/pims-6731-gbfp-prodoc_march2024_signed.pdf
118. [PDF] INVESTMENT PLAN - United Nations Development Programme https://www.undp.org/sites/g/files/zskgke326/files/2024-11/ndc_investment_plan.pdf
119. [PDF] ADAPTATION FINANCE STRATEGY GUIDELINE https://www.undp.org/sites/g/files/zskgke326/files/2026-01/afsg.pdf
120. [PDF] undp-rbap-sft-guide-2025_1.pdf https://www.undp.org/sites/g/files/zskgke326/files/2025-02/undp-rbap-sft-guide-2025_1.pdf
121. [PDF] 13TH CONSOLIDATED ANNUAL PROGRESS REPORT OF THE ... https://mptf.undp.org/sites/default/files/documents/2022-10/unredd_2021_annual_report_final_web.pdf
122. [PDF] Agricultural Research - ResearchGate https://www.researchgate.net/profile/Nihal-Agirbas/publication/366604708_Agricultural_Research_in_Nowadays/links/63aaa4d1c3c99660ebaa3b38/Agricultural-Research-in-Nowadays.pdf
123. Implication of Micronutrients in Agriculture and Health with Special ... https://www.researchgate.net/publication/256117507_Implication_of_Micronutrients_in_Agriculture_and_Health_with_Special_Reference_to_Iron_and_Zinc

124. Revisiting Sulphur—The Once Neglected Nutrient: It's Roles in Plant ... https://www.researchgate.net/publication/352999479_Revisiting_Sulphur-The_Once_Neglected_Nutrient_It's_Roles_in_Plant_Growth_Metabolism_Stress_Tolerance_and_Crop_Production
125. (PDF) Resilient and healthy food systems in low-income settings https://www.researchgate.net/publication/377485062_Resilient_and_healthy_food_systems_in_low-income_settings
126. [PDF] Advances in Plant Breeding Strategies: Agronomic, Abiotic and ... https://www.researchgate.net/profile/Daniel-Tan-21/publication/305552308_Breeding_Strategies_for_Enhanced_Plant_Tolerance_to_Heat_Stress/links/59d9de68a6fdcc2aad127c82/Breeding-Strategies-for-Enhanced-Plant-Tolerance-to-Heat-Stress.pdf
127. (PDF) Accounting for spatial trends in multi-environment diallel ... https://www.researchgate.net/publication/355470737_Accounting_for_spatial_trends_in_multi-environment_diallel_analysis_in_maize_breeding
128. Genetic Modification for Agriculture—Proposed Revision of GMO ... https://www.researchgate.net/publication/350827744_Genetic_Modification_for_Agriculture-Proposed_Revision_of_GMO_Regulation_in_Australia
129. Biofortification of baby corn: Integrating agronomic and genetic ... https://www.researchgate.net/publication/400834681_Biofortification_of_baby_corn_Integrating_agronomic_and_genetic_approaches_to_address_zinc_micronutrient_deficiencies_-_global_perspectives_and_future_challenges
130. [PDF] Revisiting Sulphur—The Once Neglected Nutrient: It's Roles in Plant ... https://www.researchgate.net/journal/Agriculture-2077-0472/publication/352999479_Revisiting_Sulphur-The_Once_Neglected_Nutrient_It's_Roles_in_Plant_Growth_Metabolism_Stress_Tolerance_and_Crop_Production/links/67dda557e2c0ea36cd87f28f/Revisiting-Sulphur-The-Once-Neglected-Nutrient-Its-Roles-in-Plant-Growth-Metabolism-Stress-Tolerance-and-Crop-Production.pdf
131. Strategies for identifying stable lentil cultivars (Lens culinaris Medik ... https://www.researchgate.net/publication/372358082_Strategies_for_identifying_stable_lentil_cultivars_Lens_culinaris_Medik_for_combating_hidden_hunger_malnourishment_and_climate_variability
132. (PDF) Using Field-Based Monitoring to Enhance the Performance of ... https://www.researchgate.net/publication/346783619_Using_Field-Based_Monitoring_to_Enhance_the_Performance_of_Rainfall_Thresholds_for_Landslide_Warning

133. Design and Validation of Low-Cost Environmental Monitoring ... https://search.proquest.com/openview/21f126013cb7a5a816c08ee03f5bf9d3/1?pq-origsite=gscholar&cbl=18750&diss=y

134. Evaluating meteorological data from weather stations, and ... - PMC https://pmc.ncbi.nlm.nih.gov/articles/PMC6024078/

135. Low cost, multiscale and multi-sensor application for flooded area ... https://www.researchgate.net/publication/325453596_Low_cost_multiscale_and_multi-sensor_application_for_flooded_area_mapping

136. Continental-scale prediction of soil organic carbon using optical ... https://www.sciencedirect.com/science/article/pii/S1161030126000559

137. (PDF) Performance of the ATMOS41 All-in-One Weather Station for ... https://www.researchgate.net/publication/348699778_Performance_of_the_ATMOS41_All-in-One_Weather_Station_for_Weather_Monitoring

138. Nanotechnology in Air Quality Management - Springer https://link.springer.com/content/pdf/10.1007/978-981-96-8339-0.pdf

139. [PDF] Advances in Engineering, Science and Technology - AWS https://uu-img.s3.ap-south-1.amazonaws.com/2021/09/Final+Copy+Proceeding+ICAEST21.pdf

140. P178292-e9050e3a-79d3-4b35-850e-12c415804397.txt https://documents1.worldbank.org/curated/en/099121824224168962/txt/P178292-e9050e3a-79d3-4b35-850e-12c415804397.txt

141. Peer reviewed papers, ICP-MS applications, ICP-QQQ Bibliography https://www.agilent.com/en/product/atomic-spectroscopy/inductively-coupled-plasma-mass-spectrometry-icp-ms/icp-ms-instruments/8900-triple-quadrupole-icp-ms/icp-qqq-bibliography?srsltid=AfmBOooxcZ93-kgYDu4ZQuCu6V62eafwlWRyUZgGi4ZxUK7f4NlN4ozH

142. Employment and the Digital Economy in the Industry 4.0/5.0 ... https://www.linkedin.com/pulse/employment-digital-economy-industry-4050-ai-era-johnny-shahinur-jo6yf

143. [PDF] Deep Learning and Computer Vision Techniques for Enhanced ... https://www.researchgate.net/publication/383689143_Deep_Learning_and_Computer_Vision_Techniques_for_Enhanced_Quality_Control_in_Manufacturing_Processes/fulltext/66d6e1b9f84dd1716c7bbf49/Deep-Learning-and-Computer-Vision-Techniques-for-Enhanced-Quality-Control-in-Manufacturing-Processes.pdf

144. PAD1146-REVISED-PAD-P149556-IDA ... - World Bank Documents https://documents1.worldbank.org/curated/en/882301468188375582/txt/PAD1146-REVISED-PAD-P149556-IDA-R2015-0130-2-Box391454B-SD-6-8-15.txt

145. [PDF] Soils, where food begins - FAO Knowledge Repository https://openknowledge.fao.org/server/api/core/bitstreams/7620bacf-90a6-4c4f-94ec-e4b14d16dde9/content

146. [PDF] Bangladesh - UNFCCC https://www4.unfccc.int/sites/SubmissionsStaging/NationalReports/Documents/198270_Bangladesh-BUR1-1-Updated%20BUR1%20Report_15_11_2023.pdf

147. Properties and Management of Acid Sulfate Soils in Southeast Asia ... https://www.researchgate.net/publication/279456060_Properties_and_Management_of_Acid_Sulfate_Soils_in_Southeast_Asia_for_Sustainable_Cultivation_of_Rice_Oil_Palm_and_Cocoa

148. Environmental-and-Social-Management-Framework.txt https://documents1.worldbank.org/curated/en/128221558340540529/txt/Environmental-and-Social-Management-Framework.txt

149. ICAR Annual Report-2017-18 - (English) PDF - Scribd https://www.scribd.com/document/412431009/ICAR-Annual-Report-2017-18-English-pdf

150. Diffuse reflectance spectroscopy (DRS) for rapid soil testing and soil ... https://doi.org/10.1111/ejss.13358

151. Envirotyping helps in better understanding the root cause of success ... https://www.frontiersin.org/journals/environmental-science/articles/10.3389/fenvs.2024.1417199/full

152. [PDF] Use of Remote Sensing to Estimate Paddy Area and Production https://www.adb.org/sites/default/files/publication/496976/remote-sensing-paddy-area-production-handbook.pdf

153. [PDF] Crop yield response to water https://www.fao.org/4/i2800e/i2800e.pdf

154. [PDF] Crop Improvement with Induced Genetic Variation to Cope with ... https://www-pub.iaea.org/MTCD/publications/PDF/CRCP-IGV-004web.pdf